

kinetic-kit

API Reference

v0.1.1 · github.com/ll-nick/kinetic-kit

Every public symbol exported by kinetic-kit, generated from the doc-comments in the source.

Everything is imported from the package root:

```
#import "@preview/kinetic-kit:0.1.1": doctoral-title-page, kit-style, outlines, thesis
```

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1 thesis

The main entry point of the package.

By default it assembles a complete doctoral thesis in the KSP-approved style, but it is fully configurable and can just as well produce a bachelor's or master's thesis. The KSP endorsement covers only doctoral theses in the default configuration; customized documents are not.

```
thesis(  
  format: str,  
  lang: str,  
  author-firstname: str,  
  author-surname: str,  
  title: content,  
  title-page: content function none,  
  front-matter: content none,  
  appendix: content none,  
  back-matter: content none,  
  serif-headings: bool,  
  heading-numbering-depth: int,  
  figure-kinds: array,  
  margin-preset: str,  
  binding-correction: length,  
  colored-links: bool,  
  draft: bool,  
  draft-info: str none,  
  doc: content  
) -> content
```

format `str`

Paper format — "a5" (148×210 mm, recommended by KSP), "17x24" (170×240 mm) or "a4" (210×297 mm). Font sizes and margins follow from it.

Default: "a5"

lang `str`

Document language — "de" or "en".

Default: "de"

author-firstname `str`

Author's first name.

Default: "Max"

author-surname `str`

Author's surname.

Default: "Mustermann"

title `content`

Thesis title.

Default: [Your Thesis Title]

title-page `content` or `function` or `none`

Title page. Pass `doctoral-title-page().with(...)` to configure the default, your own content, a function called as (title, author-firstname:, author-surname:, format:, lang:), or none.

Default: `doctoral-title-page`

front-matter `content` or `none`

Roman-numeral pages before the body — abstracts, acknowledgements, the table of contents, any page of your own, in the order written.

Default: `table-of-contents()`

appendix `content` or `none`

Appendix chapters, numbered A, A.1, ... and placed directly after the body.

Default: `none`

back-matter `content` or `none`

Pages after the appendix — list pages, bibliography, own publications, whatever the document needs, in the order written.

Default: `none`

serif-headings `bool`

Use serif headings instead of sans-serif.

Default: `false`

heading-numbering-depth `int`

Deepest heading level to number.

Default: `3`

figure-kinds `array`

Figure kinds beyond image, table and raw, as dictionaries with kind and supplement. supplement takes either one value or one per language, e.g. (de: [Algorithmus], en: [Algorithm]). Declaring a built-in kind is an error.

Default: ()

margin-preset `str`

Margin profile keyed on page count — "short" (under 200 pp), "medium" (200–399 pp) or "long" (400+ pp).

Default: "short"

binding-correction `length`

BCOR added to the inside margin for physically bound copies.

Default: 0mm

colored-links `bool`

KIT Blue hyperlinks when true, black when false.

Default: true

draft `bool`

Show the "ENTWURF"/"DRAFT" watermark on every page.

Default: false

draft-info `str` or `none`

String shown next to the watermark, e.g. a date or git SHA.

Default: none

doc `content`

Main document body.

2 doctoral-title-page

Render the doctoral thesis title page.

The default `thesis.title-page`, owning every parameter printed on the page; pass `doctoral-title-page.with(...)` to configure it. All text is always German, whatever the document language. `format`, `lang`, `title`, `author-firstname` and `author-surname` are supplied by the template when this is passed to `thesis.title-page`; setting them via `.with()` has no effect there.

```
doctoral-title-page(  
  title: content,  
  format: str,  
  lang: str,  
  author-title: str none,  
  author-firstname: str,  
  author-surname: str,  
  author-male: bool,  
  doc-degree: str,  
  doc-degree-f: str,  
  department: str,  
  university-genitive: str,  
  status-approved: bool,  
  exam-date: content str none,  
  main-advisor: content str none,  
  main-advisor-male: bool,  
  co-advisor: content str none,  
  co-advisor-male: bool  
) -> content
```

title `content`

Thesis title.

format `str`

Paper format — "a5", "17x24" or "a4". Determines font sizes and title-page margins.

Default: "a5"

lang `str`

Accepted for the title-page contract and ignored — this page is always German.

Default: "de"

author-title `str` or `none`

Academic title preceding the author's name.

Default: "M.Sc."

author-firstname str

Author's first name.

Default: "Max"

author-surname str

Author's surname.

Default: "Mustermann"

author-male bool

Selects the grammatical gender of the degree article.

Default: true

doc-degree str

Degree name in masculine form.

Default: "Doktors der Ingenieurwissenschaften (Dr.-Ing.)"

doc-degree-f str

Degree name in feminine form.

Default: "Doktorin der Ingenieurwissenschaften (Dr.-Ing.)"

department str

KIT department or faculty.

Default: "KIT-Fakultät für Maschinenbau"

university-genitive str

University name in genitive.

Default: "des Karlsruher Instituts für Technologie (KIT)"

status-approved bool

Approved version when true, submitted version when false.

Default: false

exam-date content or str or none

Date of the oral examination. Shown only when approved.

Default: `none`

main-advisor `content` or `str` or `none`

Main advisor. Shown only when approved.

Default: `none`

main-advisor-male `bool`

Selects the gendered label for the main advisor.

Default: `true`

co-advisor `content` or `str` or `none`

Co-advisor. Shown only when approved.

Default: `none`

co-advisor-male `bool`

Selects the gendered label for the co-advisor.

Default: `true`

3 Outlines

The outlines namespace holds the sections whose body the template generates: the table of contents and the back-matter list pages. Place them inside the `thesis.front-matter` / `thesis.back-matter` content of `thesis()`.

It also carries `flex-caption()`. The list pages are what switch that caption to its short form, so it does nothing without one — which is why it lives here.

```
#import "@preview/kinetic-kit:0.1.1": outlines, thesis

#show: thesis.with(
  front-matter: [
    = Kurzfassung
    ""
    #outlines.table-of-contents()
  ],
  back-matter: [
    #outlines.list-of-figures()
    #outlines.list-of("algorithm", [Algorithmenverzeichnis])
    #bibliography("refs.bib", title: [Literaturverzeichnis], style: "ieee")
  ],
)
```

3.1 table-of-contents

Print the table of contents.

```
table-of-contents(
  title: content str auto,
  depth: int
) -> content
```

title content or str or auto

Heading of the outline. auto uses Typst's localized default for the document language.

Default: auto

depth int

Deepest heading level to list.

Default: 3

3.2 list-of

Print a back-matter list page for one figure kind.

Handles the state that switches `flex-caption()` to its short form, which a hand-rolled `outline(target: ...)` would miss.

```
list-of(
  kind: function str,
  title: content str
) -> content
```

kind function or str

Figure kind to list — an element function such as image, or a string such as "algorithm".

title content or str

Heading of the list page.

3.3 list-of-figures

Print the list of figures.

```
list-of-figures(title: content str auto) -> content
```

title content or str or auto

Heading of the list page. auto uses the localized default for the document language.

Default: auto

3.4 list-of-tables

Print the list of tables.

```
list-of-tables(title: content str auto) -> content
```

title content or str or auto

Heading of the list page. auto uses the localized default for the document language.

Default: auto

3.5 list-of-listings

Print the list of listings.

```
list-of-listings(title: content str auto) -> content
```

title content or str or auto

Heading of the list page. auto uses the localized default for the document language.

Default: **auto**

3.6 flex-caption

Two-part caption: a short version for the list pages, a long one under the figure.

```
figure(image("plot.svg"), caption: outlines.flex-caption(short: [Short], long: [Long.]))
```

```
flex-caption(  
  short: content,  
  long: content  
) -> content
```

short **content**

Caption shown on the list page.

Default: **none**

long **content**

Caption shown below the figure in the document body.

Default: **none**

4 Figure Kinds

Typst keeps a separate counter and supplement for every figure kind. The template carries strings for Typst's own image, table and raw. Any other kind — pseudocode, theorems, whatever a document needs — is declared through `thesis.figure-kinds`, which supplies the caption supplement. Its list page, if you want one, is a `list-of()` call in `thesis.back-matter`.

Each entry is a dictionary with kind and supplement. supplement takes either one value or one per language:

```
#show: thesis.with(  
  figure-kinds: (  
    (kind: "algorithm", supplement: (de: [Algorithmus], en: [Algorithm])),  
    (kind: "theorem",    supplement: [Theorem]),  
  ),  
)
```

Per-chapter counter resets are automatic: they are derived from the figures present in the document, so every kind restarts each chapter whether or not it has been declared.

5 kit-style

Style constants for custom figures and components that need to match the template's visual identity: font families, sizes, line spacing, and KIT colors.

```
#import "@preview/kinetic-kit:0.1.1": kit-style

#let font-sizes = kit-style.font-sizes-by-format.at("a5")
#set text(font: kit-style.fonts.sans, size: font-sizes.small)
#rect(fill: kit-style.colors.green15, stroke: kit-style.colors.green)
```

5.1 Fonts

Field	Value	Description
fonts.serif	("Libertinus Serif",)	Body text
fonts.sans	("Libertinus Sans",)	Headings and running headers
fonts.mono	("Libertinus Mono",)	Code listings
leading	0.75em	Paragraph line spacing ($\approx 1.15\times$ at 10 pt)

5.2 Font Sizes

font-sizes-by-format is keyed by format string. Reach the sizes for one format with `kit-style.font-sizes-by-format.at(format)`. All values are length.

Field	"a5"	"17x24"	"a4"	Description
base	10pt	10pt	11pt	Body text
chapter	18pt	20pt	25pt	Chapter heading (level 1)
section	14pt	15pt	17pt	Section heading (level 2)
subsection	12pt	13pt	14pt	Subsection heading (level 3)
subsubsection	10pt	10pt	11pt	Subsubsection heading (level 4+)
title	18pt	20pt	25pt	Title page — document title
author	14pt	14pt	14pt	Title page — author name
title-info	14pt	14pt	14pt	Title page — info lines
small	8pt	8pt	9pt	Running headers, footnotes, captions
footnote	8pt	8pt	9pt	Footnote text

5.3 Colors

KIT brand colors follow a systematic naming scheme: a base name plus an optional opacity suffix (70, 50, 30, 15 = 70 %, 50 %, 30 %, 15 % tint). All values are color.

Field		Hex	Description
colors.green		#009682	KIT Green — primary brand color
colors.green70		#4CB5A7	70 % tint of colors.green
colors.green50		#7FCAC0	50 % tint of colors.green

Field		Hex	Description
colors.green30		#B2DFD9	30 % tint of colors.green
colors.green15		#D9EFEC	15 % tint of colors.green
colors.blue		#4664AA	KIT Blue — links and accents
colors.blue70		#7D92C3	70 % tint of colors.blue
colors.blue50		#A2B1D4	50 % tint of colors.blue
colors.blue30		#C7D0E5	30 % tint of colors.blue
colors.blue15		#E3E8F2	15 % tint of colors.blue
colors.black		#000000	
colors.black70		#4D4D4D	70 % tint of colors.black
colors.black50		#808080	50 % tint of colors.black
colors.black30		#B3B3B3	30 % tint of colors.black
colors.black15		#D9D9D9	15 % tint of colors.black
colors.palegreen		#82BE3C	KIT extended palette
colors.yellow		#FAE614	
colors.orange		#DCA01E	
colors.brown		#A08232	
colors.red		#A01E28	
colors.lilac		#A00078	
colors.cyanblue		#50AAE6	
colors.keyword		#0000C8	Syntax highlighting — keyword
colors.comment		#3F7F5F	Syntax highlighting — comment
colors.string		#700055	Syntax highlighting — string literal